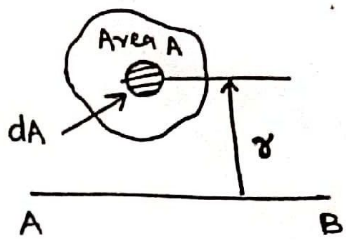
Moment of Inertia (MI)

It gives a quantitative estimate of the relative distribution of parameters like like area, mass w.r.t. some reference axis.

$$I_{AB} = \int r^2\, dA \quad \text{or} \quad \int r^2\, dm \text{, etc.}$$



Radius of Gyration :-

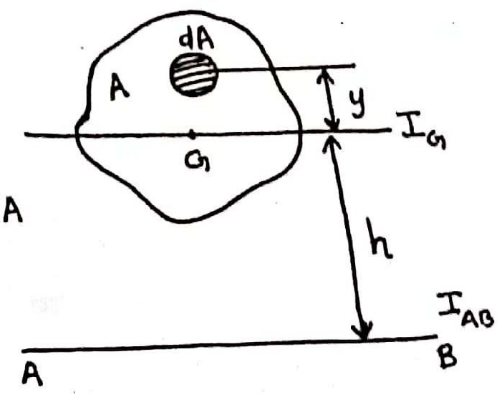
For a strip to have same moment of inertia of area 'A' w.r.t. some reference axis AB, the strip should be placed at a distance 'k' from the axis AB such that $I = Ak^2$

$$k = \sqrt{\frac{I}{A}}$$ is called radius of gyration

Parallel Axis Theorem :-

It states that if I_G is the moment of inertia of a plane lamina of area A, about its centroidal axis in the plane of the lamina, MI about any axis AB which is parallel to the centroidal axis and at a distance 'h' from the centroidal axis is given by,

$$I_{AB} = I_G + Ah^2$$



Proof :- $$I_{AB} = \int (y+h)^2\, dA$$

$$= \int y^2 dA + \int h^2 dA + 2h \int y\, dA$$

$$= I_G + h^2 A + 2hA\bar{y}$$

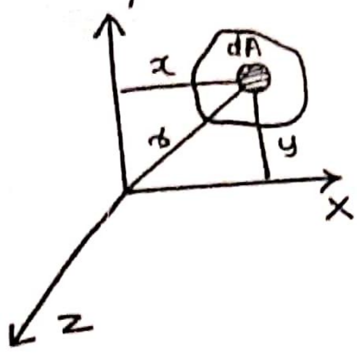
$$\therefore I_{AB} = I_G + Ah^2$$

[$\because \bar{y} = 0$ as y is measured from centroidal axis itself]

Perpendicular Axis Theorem and Polar Moment of Inertia

Moment of inertia of ZZ axis perpendicular to XX and YY axis is, $I_{zz} = I_{xx} + I_{yy}$

$I_{zz} \rightarrow$ polar moment of inertia



Proof:- $I_{zz} = \int r^2 dA$

$= \int (x^2+y^2)\, dA$

$= \int x^2 dA + \int y^2 dA$

$= I_{xx} + I_{yy}$

Product of Inertia and Principal Axes

Consider an elemental area dA of a plane area A, then the expression $\int xy\, dA$ is called the product of inertia of the area A about the x-y axes, denoted by I_{xy}.

$$I_{xy} = \int xy\, dA$$

If the product of inertia w.r.t. an axis is zero, then it is known as principal axes.

There are two principal axis perpendicular to each other. Its direction is given by,

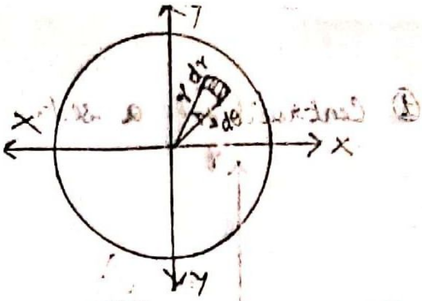
$$\tan 2\theta = \frac{2I_{xy}}{I_y - I_x}$$

MI about principal axis is known as principal MI

The maximum principal MI and minimum principal MI

$$I_{max} \text{ and } I_{min} = \frac{I_x + I_y}{2} \pm \sqrt{\left(\frac{I_x - I_y}{2}\right)^2 + (I_{xy})^2}$$

① MI of a circle about centroidal axis



Ans: MI about XX axis, $I_{xx} = \iint y^2 dA$

$$I_{xx} = \int_0^{2\pi}\int_0^R (r\sin\theta)^2 \cdot r\,d\theta \cdot dr$$

$$= \int_0^{2\pi}\int_0^R r^3\,dr \cdot \sin^2\theta\,d\theta$$

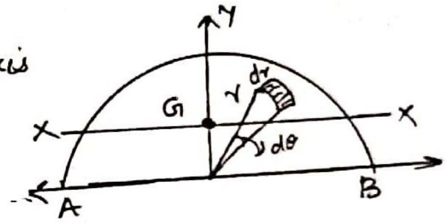
$$= \frac{R^4}{4}\int_0^{2\pi} \frac{(1-\cos 2\theta)\,d\theta}{2} = \frac{R^4}{8}\left[\theta - \frac{\sin 2\theta}{2}\right]_0^{2\pi} = \frac{R^4}{8}\left[\theta\right]_0^{2\pi}$$

$$= \frac{R^4}{8}\times 2\pi = \frac{\pi R^4}{8} = \frac{\pi D^4}{64}$$

Due to symmetry,

$I_{xx} = I_{yy} = \frac{\pi D^4}{64}$ (D – dia of circle)

② MI of semicircle about centroidal axis



MI about base AB, $I_{AB} = \iint y^2 dA$

$$I_{AB} = \int_0^{\pi}\int_0^R (r\sin\theta)^2\, r\,d\theta \times dr$$

$$= \frac{R^4}{8}\times\pi = \frac{\pi D^4}{128}$$

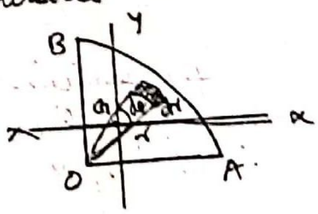
∴ about base, $I_{AB} = \frac{\pi D^4}{128}$

Using parallel axis theorem, $I_{AB} = I_G + Ah^2$

$$\therefore I_{xx} = I_G = I_{AB} - Ah^2 = \frac{\pi R^4}{8} - \frac{\pi R^2}{2}\left(\frac{4R}{3\pi}\right)^2$$

$$= R^4\left[\frac{\pi}{8} - \frac{8}{9\pi}\right] = 0.11R^4$$

③ MI of one quarter of a circle about centroidal axis



$$I_{OA} = \iint y^2 dA = \int_0^{\pi/2}\int_0^R (r\sin\theta)^2\, r\,d\theta \cdot dr$$

$$= \frac{R^4}{8}\times\frac{\pi}{2} = \frac{\pi R^4}{16} = \frac{\pi D^4}{256}$$

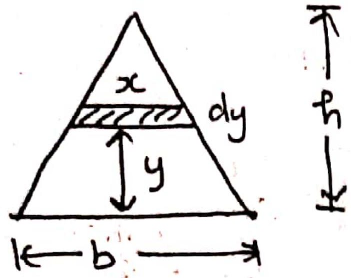
due to symmetry, $I_{OB} = I_{OA} = \frac{\pi D^4}{256}$

Using parallel axis theorem, $I_{OA} = I_{xx} + Ah^2$

$$(I_G)\, I_{xx} = \frac{\pi R^4}{16} - \frac{\pi R^2}{4}\left(\frac{4R}{3\pi}\right)^2 = R^4\left[\frac{\pi}{16} - \frac{4}{9\pi}\right] = 0.055R^4$$

Due to symmetry, $I_{xx} = I_{yy} = 0.055R^4$

Moment of Inertia of a Triangle about its base



$$I_{base} = \int y^2 \, dA.$$

By using property of triangles,

$$\frac{x}{b} = \frac{h-y}{h}$$

$$x = \frac{b}{h}(h-y)$$

$$\therefore dA = x\,dy = \frac{b}{h}(h-y)\,dy$$

$$\therefore I_{base} = \int_0^h y^2 \times \frac{b}{h}(h-y)\,dy$$

$$= \frac{b}{h}\int_0^h y^2(h-y)\,dy$$

$$= \frac{b}{h}\left[\int_0^h y^2 h\,dy - \int_0^h y^3\,dy\right]$$

$$= \frac{b}{h}\left[\left(\frac{hy^3}{3}\right)_0^h - \left(\frac{y^4}{4}\right)_0^h\right]$$

$$= \frac{b}{h}\left[\frac{h^4}{3} - \frac{h^4}{4}\right]$$

$$= \frac{b}{h}\left[\frac{h^4}{12}\right]$$

$$= \frac{bh^3}{12}$$

Virtual Work and Principle of Virtual Work

An infinitesimal displacement assumed to be given to a particle, body or to a system of bodies in equilibrium is known as virtual displacement. The product of a force and virtual displacement given to it is known as virtual work. It's imaginary work done by a true force moving through an imaginary displacement.
The principle of virtual work states that if a rigid body is in eqbm, the total work of the external forces acting on the rigid body is zero for any virtual displacement of the body consistent with the geometrical conditions of the body.

MG module 3
Cusat module 2

Plane Trusses

A plane truss is defined as a system of bars all lying in one plane and joined together at their ends in such a way as to form a rigid frame work.

A perfect truss is the truss containing least number of members required to prevent distortion of its shape when loaded. For a perfect truss,

If j is the no. of joints, 'n' no. of members = 2j - 3

If no. of members, n < 2j - 3 → deficient truss

If no. of members, n > 2j - 3 → redundant truss

A member under compression - strut
A member under tension - tie

Method of Joints

Considering each joint separately [each joint of a simple truss will be in equilibrium], applying the condition for equilibrium $\Sigma F = 0$ the forces in the members at that joint can be determined.

Method of Sections

The equilibrium of a portion of the truss is considered which is obtained by cutting the truss into two portions by an imaginary section. The condition for equilibrium $\Sigma M = 0$ is applied.

Method of Members

Trusses in which some or all the members carry loads at points other than the joints are called frames. The forces in the members of plane trusses are axial, either tensile or compressive. When the load is applied at points other than a joint, the member is subjected to bending. Here the eqbm of each member is considered separately by constructing its freebody diagram.